

Senior Research Project Approval Criteria

Before a project is approved and an advisor is assigned, your proposal must meet all the criteria below. These standards ensure that every senior project is rigorous, ethical, feasible, and college-level.

A project that does not meet these requirements will be returned for revision and will not be allowed to proceed.

1. Research Question & Purpose

Your project must have:

- ☐ A clear, testable research question or engineering goal
- ☐ A defined independent and dependent variable (for experiments)
- ☐ A measurable outcome

Not acceptable:

- Broad topics (“cancer,” “climate change,” “AI”)
- Opinion-based questions
- Projects without data or measurable outcomes

2. Scientific or Engineering Merit

Your project must:

- ☐ Be scientific, engineering-based, or research-driven
- ☐ Go beyond a literature review or simple demonstration
- ☐ Involve original data, original design, or authentic external research

Acceptable project types	Not acceptable
• Experimental (hypothesis-driven) • Engineering/ invention • University, lab, or professional research assistantship • AP Seminar & AP Research	• Surveys only • History reports • Projects that only summarize existing studies • Projects with no data collection

3. Data & Statistics Feasibility

Your design must realistically allow you to collect:

- ☐ At least 30 data points per group (control and experimental, if applicable)
- ☐ Quantitative data that can be graphed and analyzed
- ☐ Data suitable for statistical testing (t-test, ANOVA, regression, etc.)

Not acceptable

- Projects with tiny sample sizes
- Projects with only qualitative or anecdotal data
- “One-trial” experiments

4. Timeline Feasibility

Your project must be able to meet all CSW checkpoints, including:

- ☐ Hypothesis & outline (insert date here)
- ☐ Data collection
- ☐ Full draft
- ☐ Final paper

Your design must allow:

- ☐ Sufficient time for repeated trials
- ☐ Time for data analysis and writing

Projects that require long approval delays, seasonal data, or unpredictable access will not be approved unless a strong alternative plan exists.

5. Access to Materials & Mentorship

Your proposal must clearly show:

- ☐ How you will access equipment, samples, subjects, or software
- ☐ Whether you need an external mentor, lab, or facility
- ☐ A realistic plan for where and how the work will happen

If external support is required, it must be:

- ☐ Identified by the end of junior year
- ☐ Approved through the science department

6. Safety & Ethics

Your project must comply with all CSW safety policies.

You must disclose if your project involves:

- Human subjects
- Vertebrate animals
- Biohazards
- Chemicals
- Environmental or field sampling

These projects require:

- Informed consent forms
- Safety review
- Pre-approval by CSW's IRB Committee

7. Academic Integrity & Research Quality

Your proposal must be supported by:

- ☐ Scholarly sources (Google Scholar, PubMed, JSTOR, etc.)
- ☐ APA-formatted references
- ☐ Evidence that your topic is grounded in real scientific literature

Not acceptable:

- Wikipedia or general websites as sources
- AI-generated writing

- Uncited or plagiarized material

8. Scope & Level of Difficulty

Your project must be:

- ☐ Challenging and appropriate for a graduating senior
- ☐ Comparable to introductory college-level research
- ☐ Substantial enough to support:
 - ☐ A full APA paper
 - ☐ A 12-15 minute defense-style presentation (poster to reference)

Projects that are too simple, too short, or too shallow will be returned for revision.

9. Prior STEM Research Abstract

To receive project approval, all students must submit:

- ☐ A 250-300 word abstract summarizing their 9th/10th-grade STEM research project

This abstract must include:

- Research question or design goal
- Methods used
- Key results or outcomes
- What was learned

Other helpful information:

1. [Undergraduate Research Experience: A Roadmap to Guide Your Journey](#)
2. [How to Build an Independent Research Project in High School](#)
3. [The Complete Guide to Independent Research Projects for High School Students](#)